

Dirk Matthias Rudolf

Freelance TypeScript Full-Stack Engineer

Freelance since 2007

Technical Director, meta.morph Medien GmbH, 2011–2015

Diplom-Informatiker (FH)
Master of Science

Location Berlin

LinkedIn linkedin.com/in/dirk-matthias-rudolf

GitHub github.com/drudolf

Languages German (Native)
English (Fluent)

SKILLS

Web Development

TypeScript
React
Node.js / Fastify
Prisma / PostgreSQL

Smart TV Apps

Tizen
webOS
HbbTV
Video Streaming

Technical Project Management

AI Tooling / MCP
Agile Methods (Scrum, Kanban)
Test-Driven Development
Continuous Integration

PROJECTS

2023–2026

Web Development — Frontend/Backend

Built a modular tool suite for managing creatives, publishers, and targeting as part of a team, covering the full stack from database schema and API design to SPA architecture. Implemented with TypeScript, React, Node.js, Fastify, Prisma, and PostgreSQL.

Smart TV — Live TV Application

Led full technical ownership of a live TV streaming application for Tizen and webOS, from architecture through delivery. Built with TypeScript and HLS.js.

2020–2023

Smart TV — Media Library

Maintained and extended a media library app across multiple Smart TV platforms (HbbTV, Tizen, webOS, NetCast, Orsay, and others) within an agency team. Responsibilities included device-specific debugging, platform updates, feature development, and release coordination.

Smart TV — Media Library Rebuild

Rebuilt the entire stack in TypeScript, React, and Node.js, rolling out across HbbTV, Samsung (Tizen), LG (webOS), Panasonic, Philips, Sony, Foxxum, and Hisense.

Notable challenges: a codebase with platform-specific adaptations for eight manufacturers; DASH live streams with timeshift (restart) functionality; VOD, subtitle, and multi-audio track support for accessibility. Combined manual testing on physical devices with cross-platform regression tests in Jest.

2015–2020

Smart TV — Streaming Platform

Led the multi-platform port of a live streaming application to Samsung (Orsay, Tizen) and LG (NetCast, webOS) using JavaScript, Backbone.js, and DASH/HLS.

Smart TV — Streaming Platform Rebuild (Multi-Tenant)

Rebuilt as a TypeScript/React platform serving eight brands across five TV platforms (Tizen, webOS, Panasonic, Set-Top-Box, and others), developed collaboratively as a team.

Notable challenges: 5-key remote control navigation with focus management in React; unifying diverse client requirements and platform constraints into a single shared codebase; smooth video playback across all supported devices. Built with TypeScript, React, and DASH.

OPEN SOURCE

[prisma-pglite-bridge](#)

Library for fast, reliable database testing with Prisma and PGLite

Isolated Prisma database tests without Docker or an external database server — PGLite replaces PostgreSQL in-process, compatible with Vitest and Jest.

[fastify-lor-zod](#)

Type provider for the Fastify API framework with Zod schema validation

End-to-end type-safe validation of requests and responses, with automatic API documentation (OpenAPI/Swagger) from a single shared schema definition. Rewritten in TypeScript with full Zod v4 support and improved OpenAPI export.